

**TELECOM REGULATORY AUTHORITY OF INDIA**New Delhi, 23rd December, 2020

(www.trai.gov.in)

Highlights of Telecom Subscription Data as on 31st October, 2020

Particulars	Wireless	Wireline	Total (Wireless+ Wireline)
Total Telephone Subscribers (Million)	1151.81	19.99	1171.80
Net Addition in October, 2020 (Million)	3.23	-0.09	3.14
Monthly Growth Rate	0.28%	-0.43%	0.27%
Urban Telephone Subscribers (Million)	629.28	18.07	647.36
Net Addition in October, 2020 (Million)	3.12	-0.03	3.09
Monthly Growth Rate	0.50%	-0.16%	0.48%
Rural Telephone Subscribers (Million)	522.53	1.92	524.44
Net Addition in October, 2020 (Million)	0.11	-0.06	0.05
Monthly Growth Rate	0.02%	-2.96%	0.01%
Overall Tele-density*(%)	84.90%	1.47%	86.38%
Urban Tele-density*(%)	136.65%	3.92%	140.57%
Rural Tele-density*(%)	58.72%	0.22%	58.94%
Share of Urban Subscribers	54.63%	90.41%	55.24%
Share of Rural Subscribers	45.37%	9.59%	44.76%
Broadband Subscribers (Million)	713.32	21.51	734.82

❖ In the month of October, 2020, 8.80 million subscribers submitted their requests for Mobile Number Portability (MNP). With this, the cumulative MNP requests increased from 520.80 million at the end of Sep-20 to 529.60 million at the end of October-20, since implementation of MNP.

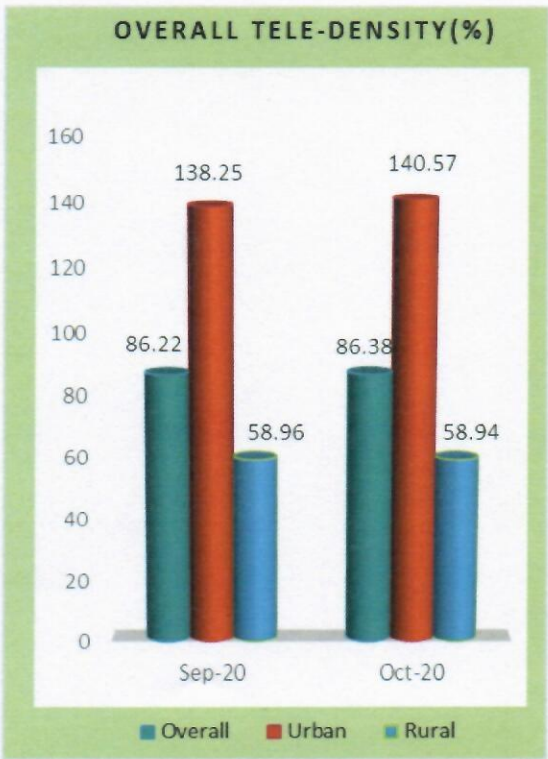
❖ Number of active wireless subscribers (on the date of peak VLR#) in October, 2020 was 960.91 million.

Note:

- Information in this Press Release is based on the data provided by the Service Providers
- * Based on the projection of population from the 'Report of the Technical Group on Population Projections for India and States 2011 – 2036', Ministry of Health & Family Welfare.
- # VLR is acronym of Visitor Location Register. The dates of peak VLR for various TSPs are different in different service areas.

I. Total Telephone Subscribers

• The number of telephone subscribers in India increased from 1,168.66 million at the end of September-20 to 1,171.80 million at the end of October-20, thereby showing a monthly growth rate of 0.27%. Urban telephone subscription increased from 644.26 million at the end of September-20 to 647.36 million at the end of October-20 and the rural subscription also increased from 524.39 million to 524.44 million during the same period. The monthly growth rates of urban and rural telephone subscription were 0.48% and 0.01% respectively during the month of October-20.



• The overall Tele-density in India increased from 86.22 at the end of September-20 to 86.38 at the end of October-20. The Urban Tele-density increased from 138.25 at the end of September-20 to 140.57 at the end of October-20. Rural Tele-density decreased from 58.96 at the end of September-20 to 58.94 at the end of October-20. The share of urban and rural subscribers in total number of telephone subscribers at the end of October-20 was 55.24% and 44.76% respectively.

Overall Tele-density (Circle/State Wise) – As on 31st October 2020



- As may be seen in the above chart, eight states have less tele-density than the all India average tele-density at the end of October-20. Delhi service area has maximum tele-density of 273.94 and the Bihar service area has minimum tele-density of 52.60 at the end of October-20.

Notes:

- Population data/projections are available state wise only.
- Tele-density figures are derived from the telephone subscriber data provided by the access service providers and the projection of population from the "Report of the Technical Group on Population Projections for India and States 2011 – 2036, Ministry of Health & Family Welfare".
- Telephone subscriber data for Delhi, includes, apart from the data for the State of Delhi, wireless subscriber data for the areas served by the local exchanges of Ghaziabad & Noida (in Uttar Pradesh) and Gurgaon & Faridabad (in Haryana).
- Data/information for West Bengal includes Kolkata, Maharashtra includes Mumbai, Tamil Nadu includes Chennai and Uttar Pradesh includes UPE & UPW service area(s).
- Data/information for Andhra Pradesh includes Telengana, Madhya Pradesh includes Chhatisgarh, Bihar includes Jharkhand, Maharashtra includes Goa, Uttar Pradesh includes Uttarakhand, West Bengal includes Sikkim and North-East includes Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland & Tripura States.

ben

II. Category-wise Growth in subscriber base

Circle Category-wise Net Additions in Telephone Subscribers in the month of October, 2020

Circle Category	Net additions in the month of October, 2020		Telephone Subscriber base as on 31 st October, 2020	
	Wireline segment	Wireless segment	Wireline segment	Wireless segment
Circle A	-37,105	2,206,776	7,743,307	394,316,385
Circle B	-6,957	545,540	4,421,140	466,345,555
Circle C	5,245	285,554	857,645	176,146,961
Metro	-48,272	189,706	6,967,755	115,000,155
All India	-87,089	3,227,576	19,989,847	1,151,809,056

Circle Category-wise monthly and yearly Growth Rates in Telephone Subscribers in the month of October, 2020

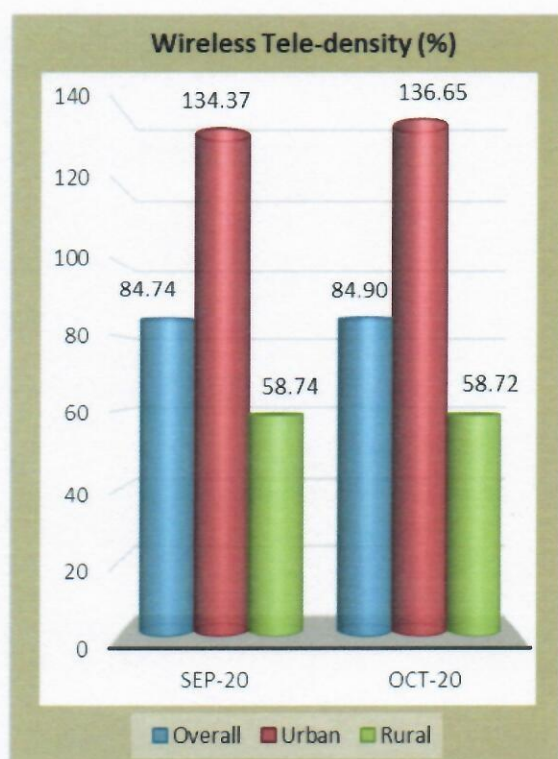
Circle Category	Monthly growth rate (%) (September-20 to October-20)		Yearly growth rate (%) (October-19 to October-20)	
	Wireline Segment	Wireless Segment	Wireline Segment	Wireless Segment
Circle A	-0.48%	0.56%	-6.66%	-2.30%
Circle B	-0.16%	0.12%	-15.06%	-3.23%
Circle C	0.62%	0.16%	0.55%	-1.24%
Metro	-0.69%	0.17%	-1.76%	-3.80%
All India	-0.43%	0.28%	-6.79%	-2.67%

Note: Circle Category-Metro includes Delhi, Mumbai and Kolkata. Data for Chennai has been included in Circle Category-A, as part of TamilNadu.

- As can be seen in the above tables, in wireless segment, on monthly basis Circles of all Categories showed growth rate in their subscribers during the month of October-20. However, on yearly basis circle of all categories have registered declining rate in their subscriber base.
- In Wireline segment, on monthly as well as on yearly basis except Metro C circle of all categories showed decline rate in their number of wireline subscribers during the month of October, 2020.

III. Wireless subscriber

- Total wireless subscribers increased from 1,148.58 million at the end of September-20 to 1,151.81 million at the end of October-20, thereby registering a monthly growth rate of 0.28%. Wireless subscription in urban areas increased from 626.16 million at the end of September-20 to 629.28 million at the end of October-20 and wireless subscription in rural areas also increased from 522.42 million to 522.53 million during the same period. Monthly growth rates of urban and rural wireless subscription were 0.50% and 0.02% respectively.

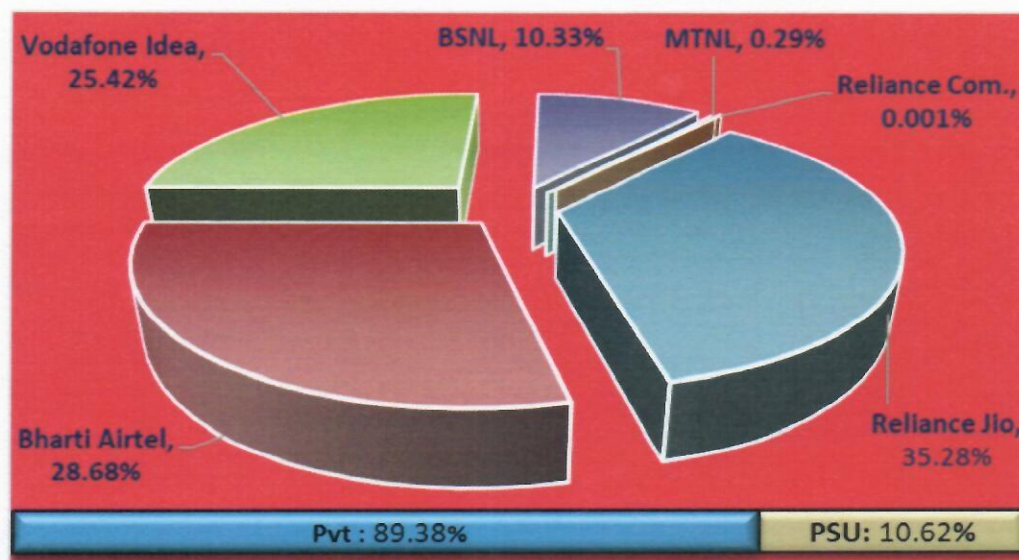


- The Wireless Tele-density in India increased from 84.74 at the end of September-20 to 84.90 at the end of October-20. The Urban Wireless Tele-density increased from 134.37 at the end of September-20 to 136.65 at the end of October-20 however, Rural Tele-density decreased from 58.74 to 58.72 during the same period. The share of urban and rural wireless subscribers in total number of wireless subscribers was 54.63% and 45.37% respectively at the end of October-20. Detailed statistics of wireless subscriber base is

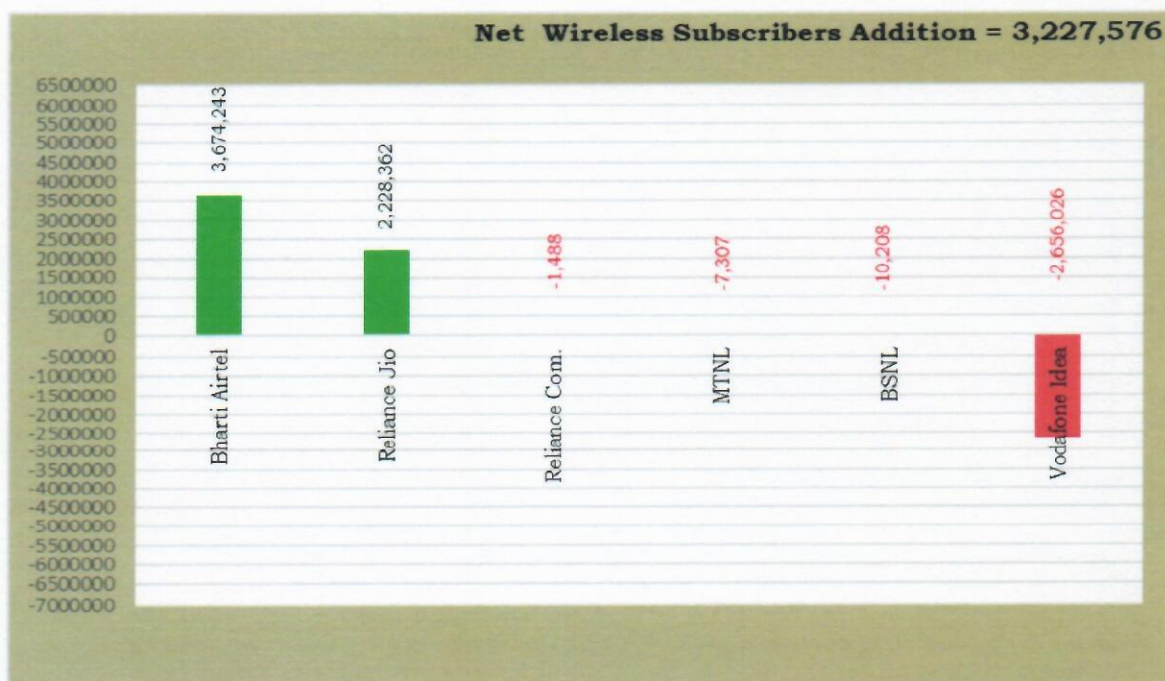
available at **Annexure-I**.

- As on 31st October, 2020, the private access service providers held 89.38% market share of the wireless subscribers whereas BSNL and MTNL, the two PSU access service providers, had a market share of only 10.62%.
- The graphical representation of access service provider-wise market share and net additions in wireless subscriber base are given below: -

Access Service Provider-wise Market Shares in term of Wireless Subscribers as on 31st October, 2020



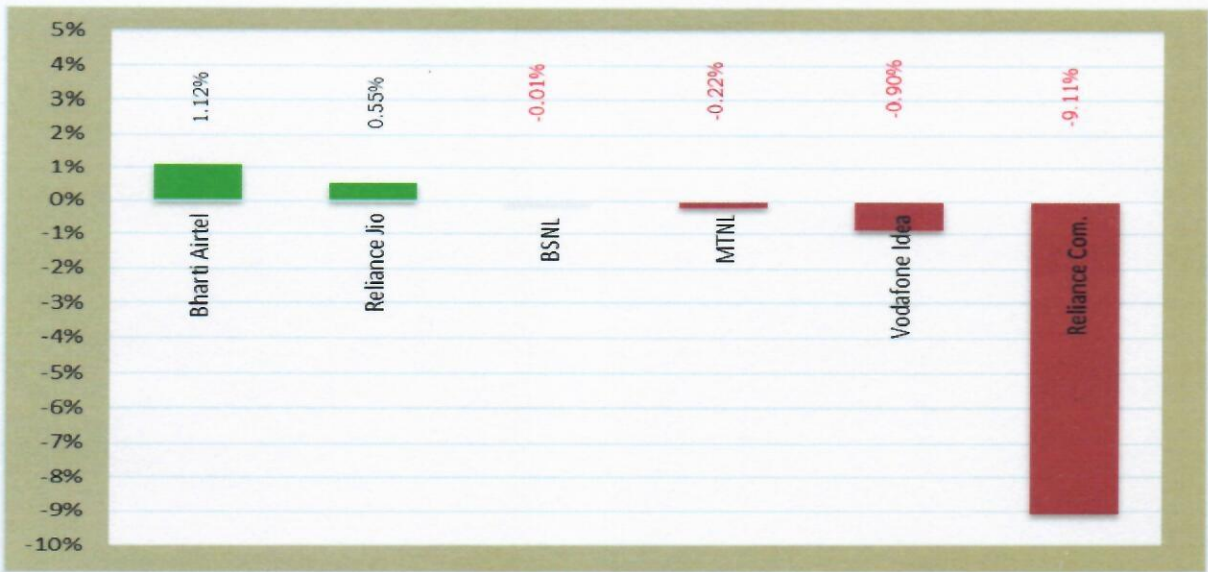
Net Additions/decline in Wireless Subscribers of Access Service Providers in the month of October 2020



Note: 1. M/s Bharti Airtel Ltd has reported the wireless subscribers including the subscribers of M/s Tata Teleservices Ltd.
 2. Virtual Network Operator (VNO) of BSNL has started reporting of its subscribers from the month of October, 2018 and the same has been included in the subscriber base of BSNL.

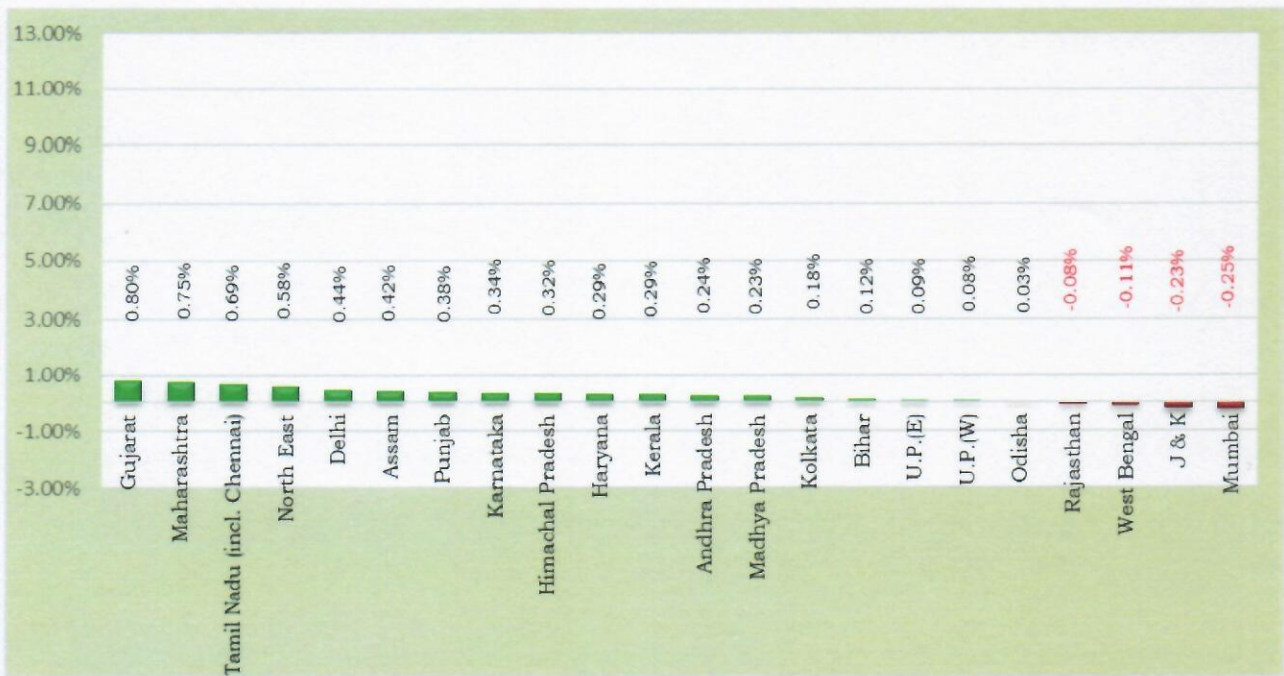
IV. Growth in Wireless Subscribers

Access Service Provider-wise Monthly Growth Rate of Wireless Subscribers in the month of October, 2020



Note: BSNL includes its Virtual Network Operator (VNO).

Service Area-wise Monthly Growth Rate of Wireless Subscribers in the month of October, 2020

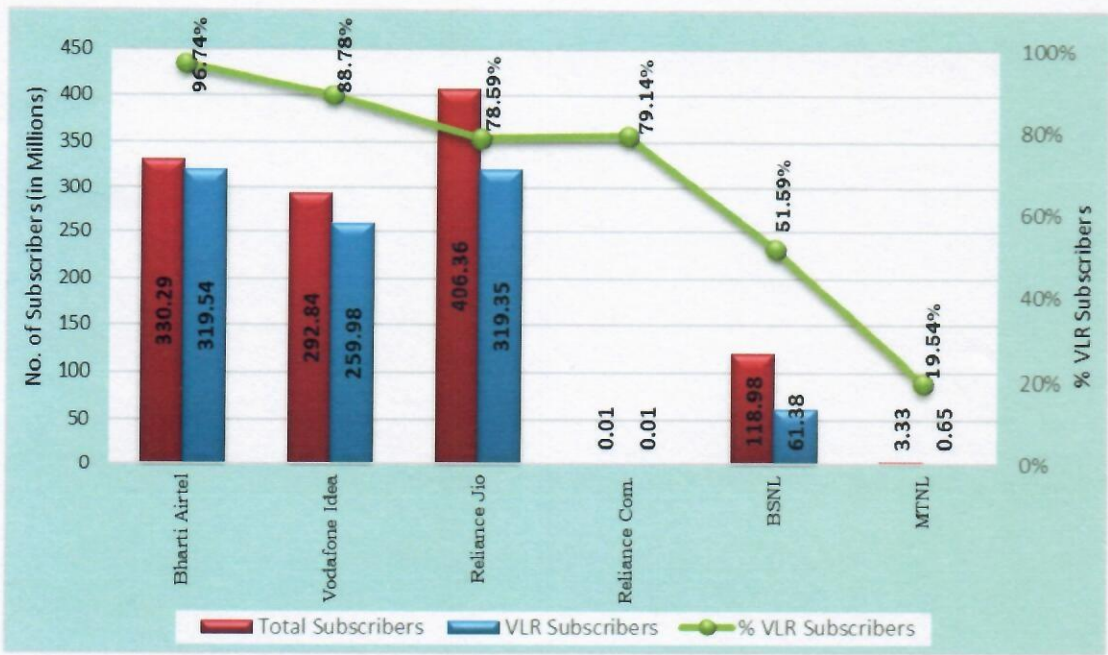


- All service areas except Rajasthan, West Bengal, J&K and Mumbai showed growth in their wireless subscribers during the month of October-20. Gujarat service area showed maximum growth of 0.80% in its wireless subscriber base during the month.

V. Active Wireless Subscribers (VLR Data)

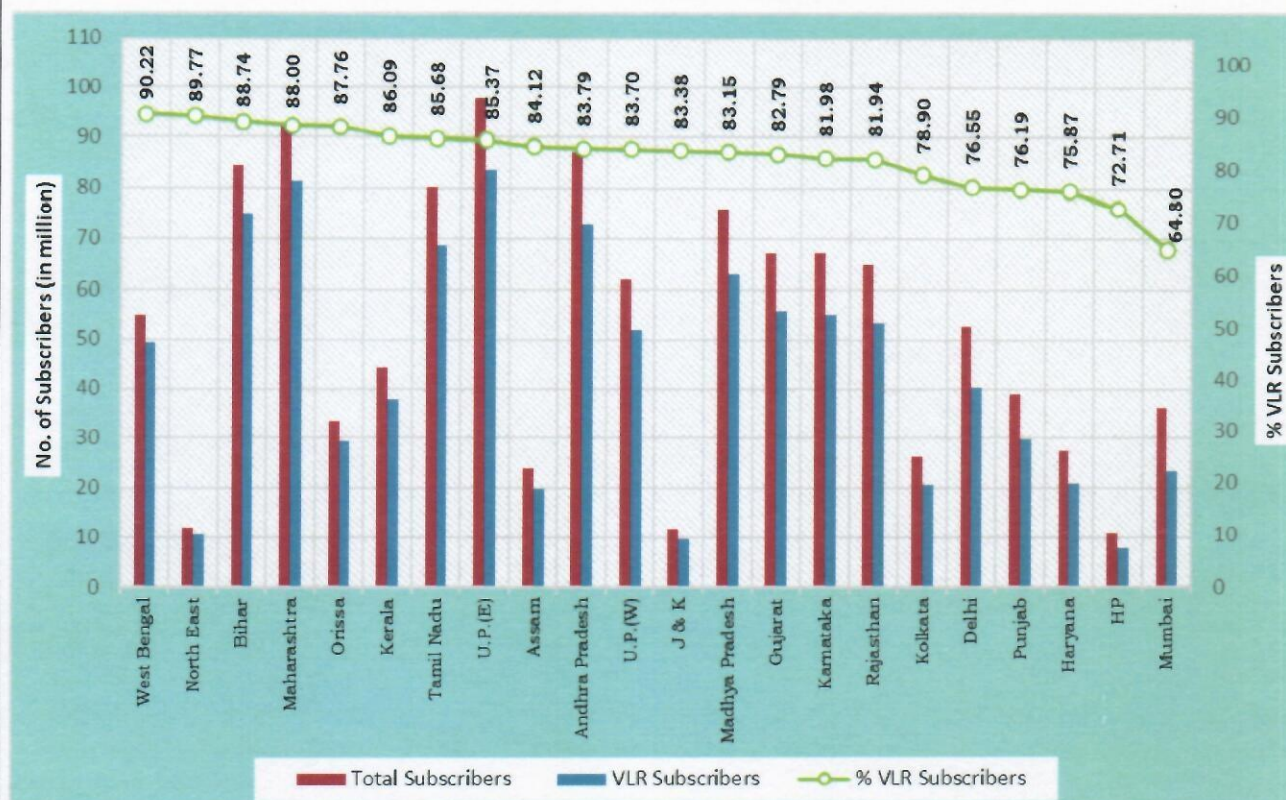
- Out of the total wireless subscribers (1,151.81 million), 960.91 million wireless subscribers were active on the date of peak VLR in the month of October-20. The proportion of active wireless subscribers was approximately 83.43% of the total wireless subscriber base.
- The detailed statistics on proportion of active wireless subscribers (also referred to as VLR subscribers) on the date of peak VLR in the month of October-20 is available at **Annexure-II** and the methodology used for reporting VLR subscribers is available at **Annexure-IV**.

Access Service Provider-wise Percentage of VLR Subscribers in the month of October 2020



- Bharti Airtel has the maximum proportion (96.74%) of its active wireless subscribers (VLR) as against its total wireless subscribers (HLR) on the date of peak VLR in the month of October-20 and MTNL has the minimum proportion of VLR (19.54%) of its HLR during the same period.

Service Area wise percentage of VLR Subscribers in the month of October 2020



VI. Mobile Number Portability (MNP)

- Intra-service area Mobile number portability (MNP) was implemented first in Haryana service area w.e.f. 25.11.2010 and in the rest of the country w.e.f. 20.01.2011. Inter-Service Area MNP has been implemented in the country w.e.f. 03.07.2015. Now, the wireless telephone subscribers can retain their mobile numbers when they relocate from one service area to another.
- During the month of October-20, a total of 8.80 million requests were received for MNP. Out of total

8.80 million new requests, 4.76 million requests received from Zone-I and 4.03 million requests received from Zone-II. The cumulative MNP requests increased from 520.80 million at the end of September-20 to 529.60 million at the end of October-20, since the implementation of MNP.

- In MNP Zone-I (Northern and Western India), the highest number of requests till date have been received in Maharashtra (about 42.53 million) followed by

Rajasthan (about 39.61 million) service area.

- In MNP Zone-II (Southern and Eastern India), the highest

number of requests till date have been received in Karnataka (about 45.81 million) followed by Andhra Pradesh (about 43.28 million).

Service Area Wise MNP Status					
Zone - 1			Zone - 2		
Service Area	Number of Porting Requests (in Million)		Service Area	Number of Porting Requests (in Million)	
	Sept-20	Oct-20		Sept-20	Oct-20
Delhi	26.29	26.66	Andhra Pradesh	42.78	43.28
Gujarat	35.02	35.70	Assam	3.89	3.93
Haryana	18.43	18.63	Bihar	22.48	23.05
Himachal Pradesh	2.53	2.56	Karnataka	45.35	45.81
Jammu & Kashmir	1.24	1.25	Kerala	13.85	14.18
Maharashtra	41.49	42.53	Kolkata	11.87	11.98
Mumbai	24.49	24.65	Madhya Pradesh	35.78	36.72
Punjab	19.81	20.07	North East	1.49	1.50
Rajasthan	39.14	39.61	Orissa	10.19	10.30
U.P.(East)	31.46	32.31	Tamil Nadu	42.30	42.82
U.P.(West)	25.25	25.93	West Bengal	25.68	26.13
Total	265.14	269.90	Total	255.66	259.70
Total (Zone-1 + Zone-2)				520.80	529.60
Net Addition (October 2020)				8.80 million	

VII. Wireline Subscribers

- Wireline subscribers decreased from 20.08 million at the end of September-20 to 19.99 million at the end of October-20. Net decrease in the wireline subscriber base was 0.09 million with a monthly decline rate of 0.43%. The share of urban and rural subscribers in total wireline

subscribers were 90.41% and 9.59% respectively at the end of October-20.

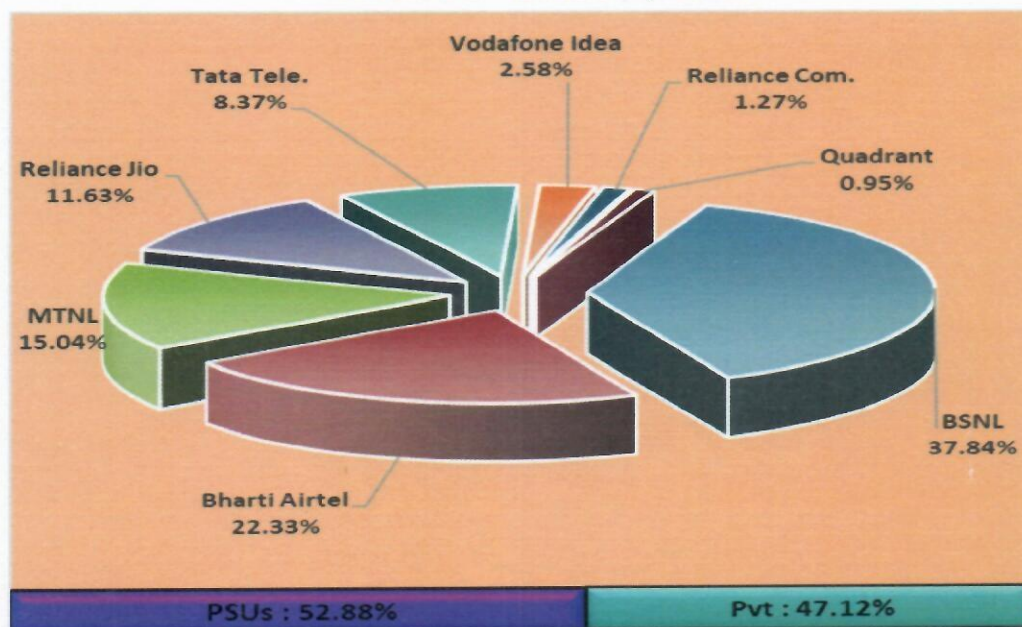
- The Overall Wireline Tele-density decreased from 1.48 at the end of September-20 to 1.47 at the end of October-20. Urban and Rural Wireline Tele-density were 3.92

and 0.22 respectively during the same period.

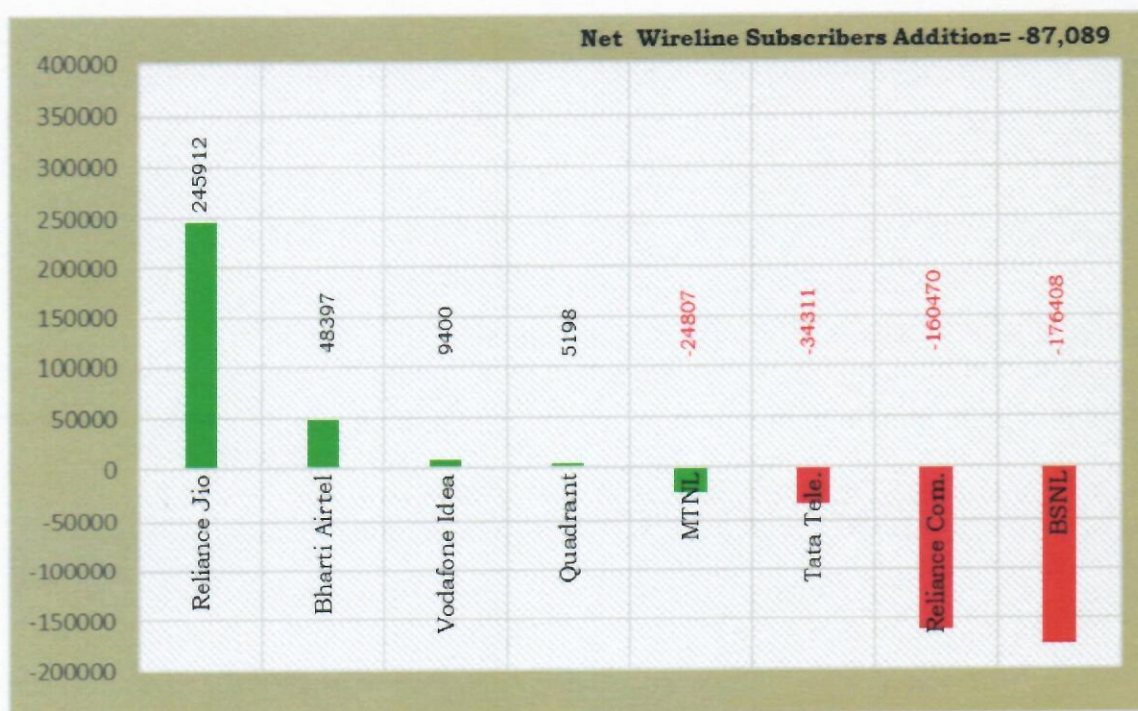
- BSNL and MTNL, the two PSU access service providers, held

52.88% of the wireline market share as on 31st October 2020. Detailed statistics of wireline subscriber base are available at **Annexure-III.**

Access Service Provider-wise Market Share of Wireline Subscribers as on 31st October, 2020



Access Service Provider-wise Net Additions in Wireline Subscribers during the month of October, 2020



[Handwritten signature]

VIII. Broadband (≥ 512 Kbps download)

- As per the reports received from 375 operators in the month of October, 2020, the number of broadband subscribers increased from 726.32 million at the end of September-20 to 734.82 million at the end of October-20 with a monthly growth rate of 1.17%. Segment-wise broadband subscribers and their monthly growth rates are as below: -

Segment-wise Broadband Subscribers and Monthly Growth Rate in the month of October, 2020

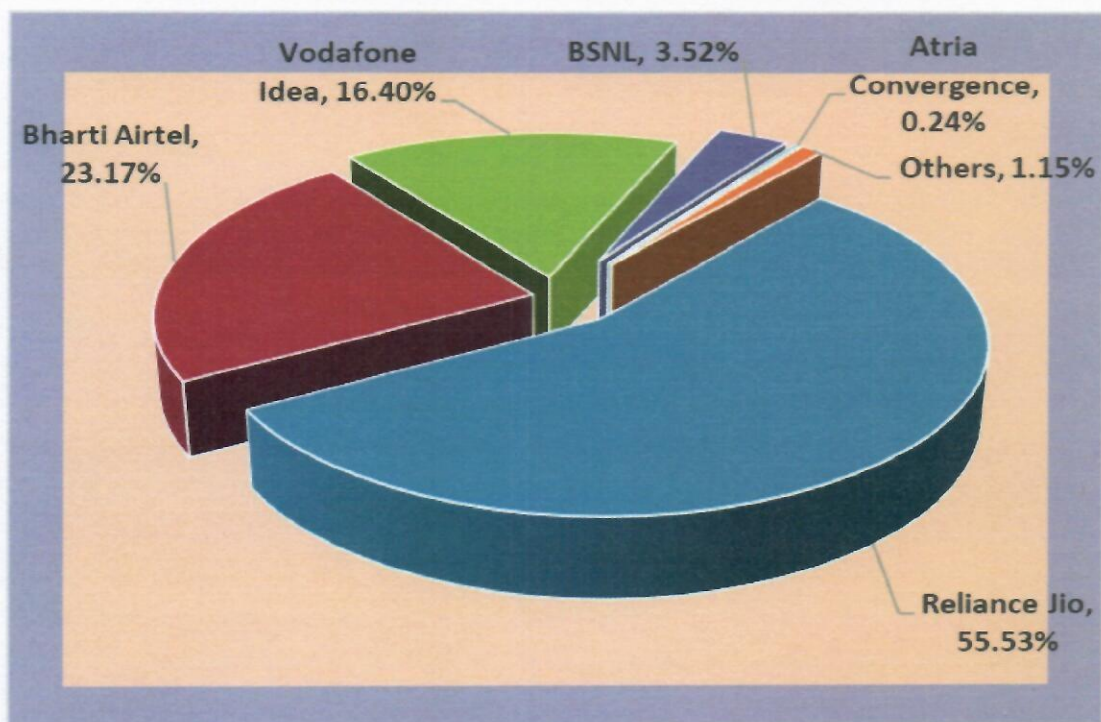
Segment	Broadband subscribers (in million)		Monthly growth rate in the month of September-20
	As on 30 th September, 2020	As on 31 st October, 2020	
Wired subscribers	21.12	21.51	1.85%
Mobile devices users (Phones and dongles)	704.57	712.67	1.15%
Fixed Wireless subscribers (Wi-Fi, Wi-Max, Point-to-Point Radio & VSAT)	0.64	0.65	1.70%
Total	726.32	734.82	1.17%

- Top five service providers constituted 98.85% market share of the total broadband subscribers at the end of October-20. These service providers were Reliance Jio Infocomm Ltd (408.06 million), Bharti Airtel (170.23 million), Vodafone Idea (120.50 million), BSNL (25.87 million) and Atria Convergence (1.74 million).

Note: Some wireless service providers exclude incidental data users from their subscriber base, based on minimum usage decided by them.

- The graphical representation of the service provider-wise market share of broadband services is given below: -

**Service Provider-wise Market Share of Broadband
(wired + wireless) Services as on 31st October 2020**



- As on 31st October, 2020, the top five Wired Broadband Service providers were BSNL (7.75 million), Bharti Airtel (2.67 million), Atria Convergence Technologies (1.74 million), Reliance Jio Infocomm Ltd (1.70 million) and Hathway Cable & Datacom (1.05 million).
- As on 31st October, 2020, the top five Wireless Broadband Service providers were Reliance Jio Infocom Ltd (406.36 million), Bharti Airtel (167.56 million), Vodafone Idea (120.49 million), BSNL (18.12 million) and Tikona Infinet Ltd. (0.31 million).

Contact details in case of any clarification: -

Shri S. K. Mishra, Pr. Advisor (F&EA),
Telecom Regulatory Authority of India
Mahanagar Doorsanchar Bhawan
Jawahar Lal Nehru Marg,
New Delhi – 110002,
Ph: 011-23221856
Fax: 011-23235249
E-mail: skmishra.trai@nic.in


(S. K. Gupta)
Secretary, TRAI

Wireless Subscriber Base																	Annexure-I	
Circle	Bharti Airtel (including Tata Tele.)		Reliance Com.		Vodafone Idea		BSNL		BSNL (VNO's)		MTNL		Reliance Jio		Total		Net Addition	
	Sep-20	Oct-20	Sep-20	Oct-20	Sep-20	Oct-20	Sep-20	Oct-20	Sep-20	Oct-20	Sep-20	Oct-20	Sep-20	Oct-20	Sep-20	Oct-20		
Andhra Pradesh	29367090	29668196	1694	1503	16630879	16528248	9781494	9713693					31184600	31262744	86965757	87174384	208627	
Assam	8485892	8591982			3581363	3500718	2888111	2911665					8684391	8734549	23639757	23738914	99157	
Bihar	34560614	34712441	351	299	12862375	12646068	5875219	5897628					30983400	31123759	84281959	84380195	98236	
Delhi	15477754	15587303	1190	1098	16101122	16124925					2174003	2172229	18740188	18841531	52494257	52727086	232829	
Gujarat	10372247	10796568	614	451	25474201	25393117	6109492	6112210					24830736	25021618	66787290	67323964	536674	
Haryana	4767758	4856620	130	118	8088541	8075479	5073981	5068181					9429315	9439947	27359725	27440345	80620	
Himachal Pradesh	3280745	3282662	104	102	744572	732036	2988503	2997054					3894963	3932415	10908887	10944269	35382	
J & K	5558818	5530091	0	0	540719	530354	1254948	1259888					4374426	4381138	11728911	11701471	-27440	
Karnataka	28750088	29038010	2269	2293	10266022	10147702	7256391	7256922					20785830	20844267	67060600	67289194	228594	
Kerala	6014889	6138044	698	652	17362961	17319382	10951796	10955911					10653841	10698091	26111702	26157542	45840	
Kolkata	5945343	5935061	33	30	7185220	7129319	2327265	2395041					9723173	9765595	44053517	44179584	126067	
Madhya Pradesh	14051370	14241776	791	765	23083055	22951010	6303315	6295680					32346106	32471133	75784637	75960364	175727	
Maharashtra	16390350	16806314	859	703	34017639	33712109	7083722	7040134					34256776	34874588	91749346	92433848	684502	
Mumbai	9486228	9587522	1651	1528	11239051	10964403					1163395	1157862	14314165	14404212	36204490	36115527	-88963	
North East	5189422	5220598	0	0	1535017	1517824	1367606	1387154					3902705	3939263	11994750	12064839	70089	
Odisha	10842112	10834583	386	371	2697878	2676778	6357676	6324069					13409091	13481472	33307143	33317273	10130	
Punjab	10298938	10395236	402	392	8677625	8656560	5856662	5885579					13991923	14035256	38825550	38973023	147473	
Rajasthan	21261002	21336223	416	386	12646298	12458919	6356489	6345382					24895132	24965553	65159337	65106463	-52874	
Tamil Nadu (incl. Chennai)	25191670	25422052	2512	1930	19228919	19268866	10921003	11108090	92482	92489			24110030	24201568	79546616	80094995	548379	
U.P.(E)	31790706	32388360	737	717	24752673	24129613	11675292	11625974					29555077	29714290	97774485	97858954	84469	
U.P.(W)	14718017	15092973	340	311	21041049	20773006	6018620	5887605					20180022	20250897	61958048	62004792	46744	
West Bengal	14810655	14823336	1157	1197	17738564	17603281	2450590	2420100					19883750	19974116	54884716	54822030	-62686	
Total	326611708	330285951	16334	14846	295495743	292839717	118898175	118887960	92482	92489	3337398	3330091	404129640	406358002	1148581480	1151809056	3227576	
Net Addition		3674243		-1488		-2656026		-10215		7		-7307		2228362	0	3227576		
Rural Subscribers	153842506	156020439	0	0	160374597	157339002	38082539	37934237	0	0	45437	45400	170073110	171187748	522418189	522526826	108637	

Note: 1. The subscribers of M/s Bharti Airtel includes subscribers of M/s Tata Teleservices Ltd.

2. Virtual Network Operator (VNO) of BSNL has reported its subscriber number from the month of October, 2018.

Note: 1. The subscribers of M/s Bharti Airtel includes subscribers of M/s Tata Teleservices Ltd.

2. Virtual Network Operator (VNO) of BSNL has reported its subscriber number from the month of October, 2018.

Proportion of VLR on the date of Peak VLR in the month of October-20 (%) Annexure-II

Circle	Bharti	BSNL	Vodafone Idea	M TNL	Reliance Com.	Reliance Jio	Total
Andhra Pradesh	98.50	57.95	94.64		80.04	72.13	83.79
Assam	98.57	49.30	88.61		-	79.72	84.12
Bihar	93.91	46.07	85.91		151.18	92.21	88.74
Delhi	89.30		76.38	11.84	35.92	73.61	76.55
Gujarat	100.99	46.59	91.31		35.92	75.15	82.79
Haryana	110.12	36.02	90.45		54.24	67.18	75.87
Himachal Pradesh	102.53	36.67	101.12		32.35	69.99	72.71
J & K	95.95	57.26	83.92		-	74.97	83.38
Karnataka	93.35	52.15	86.57		64.02	74.28	81.98
Kerala	100.67	65.90	98.41		37.88	77.75	86.09
Kolkata	93.57	44.14	86.58		-	73.43	78.90
Madhya Pradesh	97.30	46.02	81.42		47.32	85.37	83.15
Maharashtra	103.40	55.67	92.33		67.99	82.92	88.00
Mumbai	71.66		76.85	33.99	193.72	53.53	64.80
North East	100.34	64.46	87.47		-	85.56	89.77
Orissa	102.19	62.48	90.77		17.79	87.43	87.76
Punjab	99.67	39.57	89.55		36.22	65.93	76.19
Rajasthan	93.27	44.91	90.92		43.26	77.18	81.94
Tamil Nadu (incl. Chennai)	96.53	67.88	90.93		116.27	78.33	85.68
U.P. (E)	99.09	36.34	90.21		26.92	85.67	85.37
U.P. (W)	104.78	42.67	87.75		10.93	75.77	83.70
West Bengal	97.38	77.06	89.69		16.54	86.96	90.22
Total	96.74	51.59	88.78	19.54	79.14	78.59	83.43

Note: Peak VLR figures in some circles of some of the service providers are more than their HLR figures due to a large number of inroamers.

Wireline Subscriber Base															Annexure-III				
Service Area	BSNL		MTNL		Bharti Airtel		Reliance Com.		Tata Tele.		Quadrant		Vodafone Idea		Reliance Jio		Total		Net Addition
	Sep-20	Oct-20	Sep-20	Oct-20	Sep-20	Oct-20	Sep-20	Oct-20	Sep-20	Oct-20	Sep-20	Oct-20	Sep-20	Oct-20	Sep-20	Oct-20	Sep-20	Oct-20	
Andhra Pradesh	671604	652610			228534	232544	24265	25126	162180	156446			62290	62160	204571	231459	1353444	1360345	6901
Assam	88179	86873											3480	3450	25245	29204	116904	119527	2623
Bihar	136451	135936					1064	19	7619	7528			1830	1830	44948	53564	191912	198877	6965
Delhi	0	0	1361213	1352768	1453257	1470770	54847	42830	145610	144248			76820	75595	172397	192862	3264144	3279073	14929
Gujarat	543509	532431			114715	115073	7003	5540	83005	82006			32907	33387	180740	197976	961879	966413	4534
Haryana	151169	147400			28306	26973	895	3	36795	36605			360	330	73457	81231	290982	292542	1560
Himachal Pradesh	85059	83470			0	0	684	3	1839	1779			30	30	4745	6109	92357	91391	-966
J & K	102635	100586			9585	0	0	0	0	0			0	0	43115	50223	155335	150809	-4526
Karnataka	816343	796592			783151	793102	89354	29901	263744	262385			90212	101792	140128	172767	2182932	2156539	-26393
Kerala	1279325	1215503			60921	62307	8062	6483	19620	19443			6150	6090	35834	42814	1409912	1352640	-57272
Kolkata	357271	348775			138089	138197	31930	11751	47898	47322			13130	13250	93064	101811	681382	661106	-20276
Madhya Pradesh	288195	283062			249866	254690	3511	5	11785	11998			3675	3585	102763	118478	659795	671818	12023
Maharashtra	834521	837260			125867	130545	23558	13106	241556	236645			30373	29203	65164	71053	1321039	1317812	-3227
Mumbai	0	0	1669653	1653291	374374	379755	109083	81190	517936	501531			93581	93486	305874	318323	3070501	3027576	-42925
North East	76059	75262											270	270	17154	20193	93483	95725	2242
Odisha	162307	159713					1519	6	9283	8145			6240	6210	23060	27242	202409	201316	-1093
Punjab	268507	260831			138258	139629	6169	2575	12379	12237	185615	190813	3210	3240	69937	74841	684075	684166	91
Rajasthan	299894	288741			65473	66817	8826	5622	11496	11585			16430	16460	74466	81998	476585	471223	-5362
Tamil Nadu (incl. Chennai)	1044710	1008796			535940	543177	37998	28936	120865	119423			39165	39635	182440	202231	1961118	1942198	-18920
U.P.(E)	185049	181230			71477	70649	1792	28	7727	7747			19960	20020	80426	89949	366431	369623	3192
U.P.(W)	171597	167237			36750	38732	1640	10	4534	4515			5430	4920	113179	129339	333130	344753	11623
West Bengal	177982	201650					1418	14	2439	2411			120	120	25228	30180	207187	234375	27188
Total	7740366	7563953	3030866	3006059	4414563	4462960	413618	253148	1708310	1673999	185615	190813	505663	515063	2077935	2323847	20076936	19989847	-87089
Net Addition				-24807		48397		-160470		-34311		5198		9400		245912		-87089	
Rural Subscribers	1892625	1833572	0	0	0	0	651	0	42862	41987	32169	31177	0	0	7042	10058	1975349	1916794	-58555

VLR Subscribers in the Wireless Segment

Home Location Register (HLR) is a central database that contains details of each mobile phone subscriber that is authorized to use the GSM core network. The HLRs store details of every SIM card issued by the service provider. Each SIM has a unique identifier called an International Mobile Subscriber Identity (IMSI), which is the primary key to each HLR record. The HLR data is stored for as long as a subscriber remains with the service provider. HLR also manages the mobility of subscribers by means of updating their position in administrative areas. It sends the subscriber data to a Visitor Location Register (VLR).

Subscriber numbers reported by the service providers is the difference between the numbers of IMSI registered in service provider’s HLR and sum of other figures as given below: -

1	Total IMSI's in HLR (A)
2	Less: (B = a + b + c + d + e)
a.	Test/Service Cards
b.	Employees
c.	Stock in hand/in Distribution Channels (Active Card)
d.	Subscriber Retention period expired
e.	Service suspended pending disconnection
3	Subscribers Base (A-B)

Visitor Location Register (VLR) is a temporary database of the subscribers who have roamed into the particular area, which it serves. Each base station in the network is served by exactly one VLR; hence a subscriber cannot be present in more than one VLR at a time.

If subscriber is in active stage i.e. he is able to send/receive calls/SMSs he is available both in HLR and VLR. However, it may be possible that the subscriber is registered in HLR but not in VLR due to the reason that he is either switched-off or moved out of coverage area, not reachable etc. In such circumstances he will be available in HLR but not in VLR. This causes difference between subscriber number reported by the service providers based on HLR and numbers available in VLR.

The VLR subscriber data calculated here is based on active subscribers in VLR on the date of Peak subscriber number in VLR of the particular month for which the data is being collected. This data is to be taken from the switches having the purge time of not more than 72 hours.

